

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY

PAPER 2

9189/2

JUNE 2013 SESSION

1 hour 15 minutes

Candidates answer on the question paper.

Additional materials:

Data Booklet

Mathematical Tables and/or Calculator

TIME 1 hour 15 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces at the top of this page.

Answer all questions.

Write your answers in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE

1	
2	
3	
4	
5	
TOTAL	

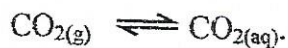
This question paper consists of 10 printed pages and 2 blank pages.

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- 1 (a) Define the term *dynamic equilibrium*.

[1]

- (b) Carbon dioxide is used as an effervescent in fizzy drinks. The equilibrium involved is represented as



Explain why

- (i) bubbles of CO_2 gas are released when a fizzy drink is poured into a glass,

[1]

- (ii) carbon dioxide is dissolved in the drink at high pressure.

[1]

- (c) Hydrogen iodide decomposes according to the equation



Describe how the rate of reaction can be determined by colorimetric analysis.

[3]

(d) Describe the effect of these factors on the rate of decomposition of HI.

(i) increasing pressure

[1]

(ii) using a catalyst

[2]

(iii) decreasing temperature

[2]

[Total:11]

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- 2 (a) (i) Define the term *standard electrode potential*.

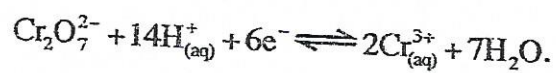
_____ [1]

- (ii) Give two reasons why a platinised platinum metal is used in preference to other metals in a half cell where the reaction does not involve a metal element.

1. _____

2. _____
_____ [2]

- (b) Show with the aid of a labelled diagram how the E^θ value of the half cell represented by this equation is measured.



- (c) State the limitation of the E^θ value approach. [3]

_____ [1]
[Total: 7]

3

- (a) Fluorine reacts with hydrogen at -200°C in the dark whereas chlorine reacts slowly with hydrogen in the dark and explodes in bright sunlight.

- (i) Use information from the data booklet to explain the greater reactivity of fluorine,

[2]

- (ii) Describe how iodine will react with hydrogen. Illustrate your answer by using information from the data booklet.

[2]

- (b) (i) State the trend in the oxidising ability of the halogens.

[1]

- (ii) Explain the trend in the oxidising ability of the halogens.

[2]

- (c) Describe and explain the variation in volatility of the halogens down the group.

[2]

(d) Eutrophication is an environmental consequence of using nitrate fertilisers.

(i) State **two** effects of eutrophication to humans?

1. _____

2. _____

[2]

(ii) Suggest ways of reducing the problem of eutrophication.

_____ [2]

(e) Nitrogen undergoes few reactions.

Explain the lack of reactivity of nitrogen.

_____ [1]

[Total: 14]

- 4 (a) Fig. 1 shows the structure of morphine, a powerful painkiller, isolated from the opium poppy tree.

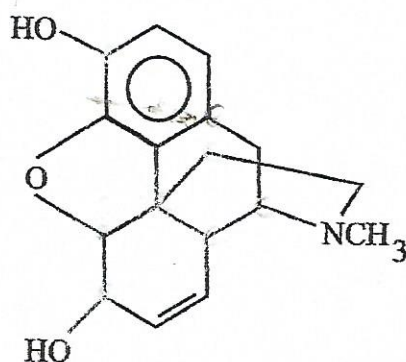
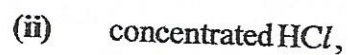
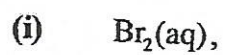


Fig. 1

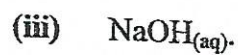
- (i) State the functional group(s) present in morphine.
- _____
- _____
- [2]
- (ii) Indicate on Fig. 1 by means of asterisks (*) the chiral centres in this molecule.
- [1]

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- (b) Draw the structural formulae of the organic products formed when morphine reacts with



[1]



[2]

[1]

- (c) State the type of reaction in b(iii)

[1]

[Total: 8]

- 5 (a) Fig. 2 shows the structure of salbutamol, a drug used in ventolin inhalers to help asthma sufferers breathe more easily.

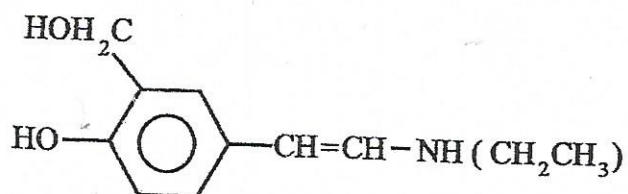


Fig. 2

- (i) State the observation made when salbutamol is treated with

1. aqueous bromine,

[1]

2. acidified potassium dichromate,

[1]

3. phosphorus pentachloride.

[1]

- (ii) Draw the structure of the organic product(s) formed when salbutamol reacts with

1. ethanolic NaOH,

[1]

2. $\text{CH}_3\text{CH}_2\text{Br}$,

[1]

3. ethanoyl chloride,

[1]

- (b) State any two uses of esters.

[2]

[Total: 8]